TARGET CAT 2025

MBA FASTRACK

QUANT : Arithmetic

Time Speed Distance III
& Averages

Lecture No.- 09

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Recap of Previous Lecture :

$$\boxed{s = \frac{d}{t}}$$

$$\boxed{t = \frac{d}{s}}$$

$$\boxed{d = s \times t}$$

$1\ km/h = \frac{5}{18}\ m/s$

$1\ m/s = \frac{18}{5}\ km/h$

$1\ min = \frac{1}{60}\ hr$

$\frac{s_1}{s_2} = \frac{t_2}{t_1}$

$\boxed{\because d\ \text{constant}}$

$\frac{s_1}{s_2} = \frac{2}{3}$

$\frac{t_1}{t_2} = \frac{3}{2}$

$\boxed{\because t\ \text{constant}}$

$\frac{s_1}{s_2} = \frac{2}{3}$

$\frac{s_1}{s_2} = \frac{d_1}{d_2}$

$\frac{d_1}{d_2} = \frac{2}{3}$

$\boxed{\because s\ \text{constant}}$

$\frac{d_1}{d_2} = \frac{2}{3}$

$\frac{t_1}{t_2} = \frac{2}{3}$

Recap of Previous Lecture :

- Trains
- Boats
- Escalator

$\boxed{\text{Relative Speed}}$

$\boxed{S_1 > S_2}$

$\boxed{S_R = S_1 - S_2} \longrightarrow$ Same

$\boxed{S_R = S_1 + S_2} \longrightarrow$ oppo.

TOPICS *to be covered*

1) Concepts & Numerical based on 'Linear Races'

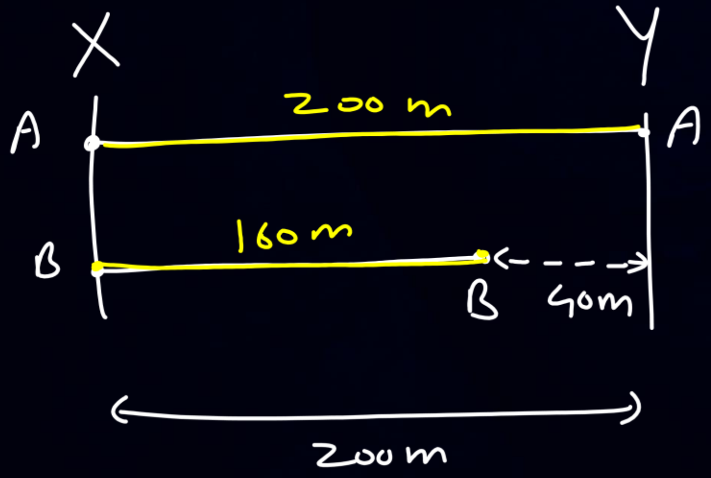
2) Concepts & Numerical based on 'Circular Races'

3) Concept & Numerical based on 'Averages'

4) Problems for Practice & CAT PYQs

Linear Races :

'A' beats 'B' by 40 m in the race of 200 m



'A' gives 'B' a Start of 20 'm'

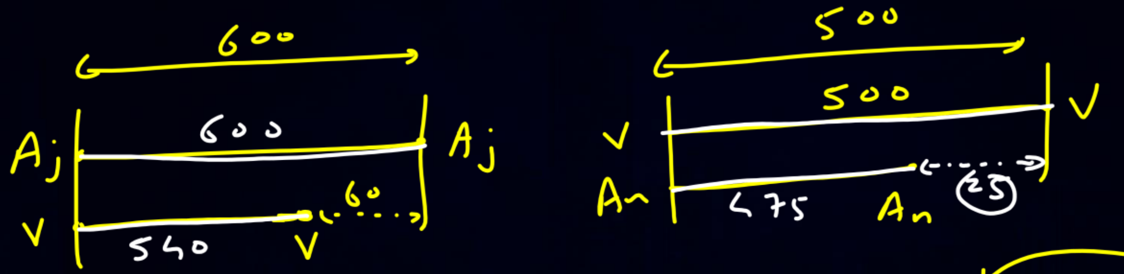
Dead-heat

↓

Tie

QUESTION- 1

#Q. In a race of 600 m, Ajay beats Vijay by 60 meters and in a race of 500 m, Vijay beats Anjan by 25 meters. By how many meters will Ajay beat Anjan in a 400 meters race?



Aj : V
~~600~~ : ~~540~~
$\boxed{10 : 9}$

V : An
~~500~~ : ~~475~~
$\boxed{20 : 19}$

Aj : V : An
10 : 9 → 9
20 ← 20 : 19
200 : 180 : 171
29 m

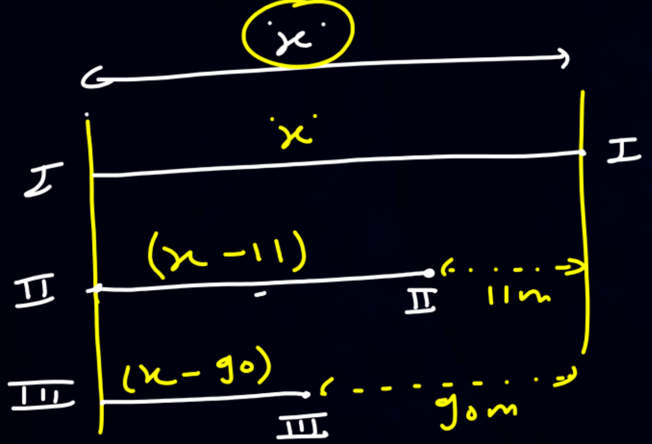
200 → 29
400 → $\boxed{58}$ x

$\cancel{200}x = \cancel{400} \times 29$
$\boxed{x = 58}$

- (A) 48 m
- (B) 52 m
- (C) 56 m
- (D) 58 m

QUESTION- 2

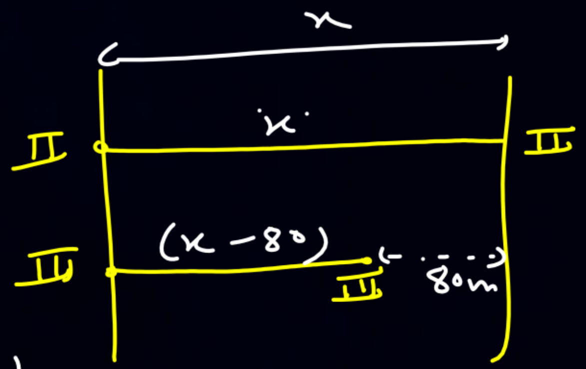
#Q. In a race of three horses, the first beat the second by 11 metres and the third by 90 metres. If the second beat the third by 80 metres, what was the length, in metres, of the racecourse? $\boxed{880}$ [CAT 2019 : Slot 1]



~~I~~ : II : III

~~x~~ : (x − 11) : (x − 90)

x : (x − 80)



$$(x-11)(x-80) = x(x-90)$$

$$\cancel{x^2} - 80x - 11x + 880 = \cancel{x^2} - 90x$$

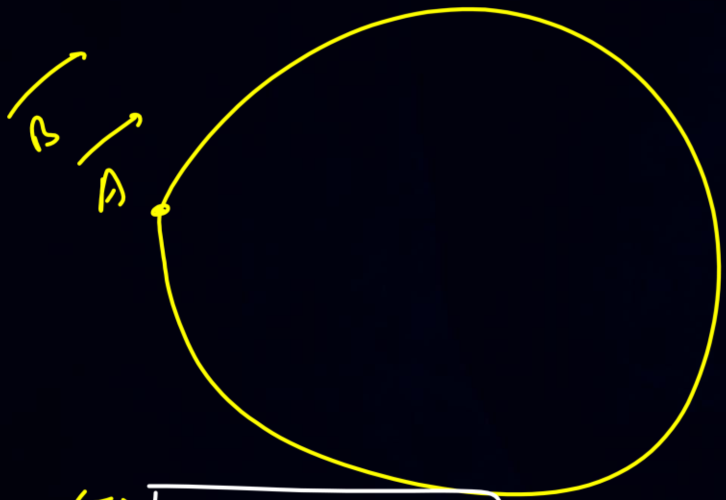
$$-91x + 880 = -90x$$

$$\boxed{880} = 91x - 90x = x$$

Circular Races :

$\boxed{L = 300\,m}$

$\boxed{\text{Same dir}^n}$



(I) $\boxed{A = 50\,m/s}$

(II) $\boxed{B = 60\,m/s}$

$T_1 = \frac{300}{50} = 6$ sec

$T_2 = \frac{300}{60} = 5$ sec

$A \rightarrow 6, 12, 18, 24, 30, 36 - - - .$

$B \rightarrow 5, 10, 15, 20, 25, 30 - - -$

When will 'A' & 'B' meet for the first time at starting point

$= \boxed{\text{LCM of } (T_1 \,\&\, T_2)}$

$= \text{LCM of} (6, 5) = 30$ sec

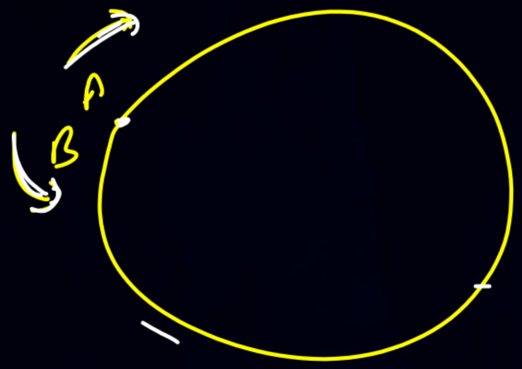
$T_1 \rightarrow$ time required for I^{st} runner to complete one round

$T_2 \rightarrow$ ______ ______ II^{nd} runner to complete one round

Circular Races :

$L = 300m$

$\boxed{\text{opposite dir}^n}$



When will they meet for the first time at $\boxed{\text{Starting Point}}$

$= \text{LCM of } (T_1 \,\&\, T_2)$

$= \text{LCM of } (6 \,\&\, 5) = 30 \text{ sec}$

(I) $A = 50$ m/s

(II) $B = 60$ m/s

$T_I = \frac{300}{50} = 6$ sec

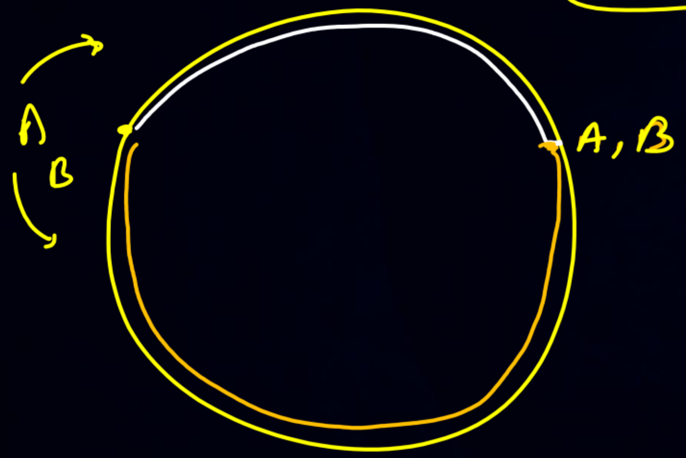
$T_{II} = \frac{300}{60} = 5$ sec

$A \rightarrow 6, 12, 18, 24, 30 \ldots$

$B = 5, 10, 15, 20, 25, 30 \ldots$

Circular Races :

oppo. dirⁿ



When will they meet for the first time anywhere on the track

dist → L

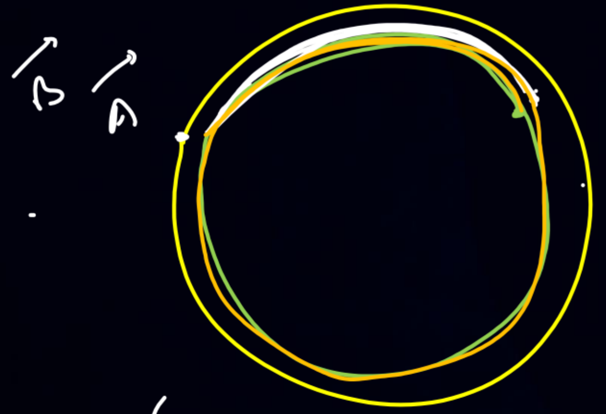
Speed → $S_A + S_B$

$$\text{Time} = \frac{\text{dist}}{\text{Speed}}$$

$$\boxed{t = \frac{L}{S_A + S_B}}$$

Circular Races :

$\boxed{\text{Same dirn}}$



$S_A > S_B$

$A = 20$

$B = 80$

20 60

When will they meet for the first time (anywhere on the track)

dist → L

Speed → $S_A - S_B$

$$\boxed{t = \frac{L}{S_A - S_B}}$$

$\boxed{\text{Meeting at SP}}$

$\boxed{\text{Same dirn}}$ or $\boxed{\text{oppo dirn}}$

$t \rightarrow \text{Lcm}(T_1 \,\&\, T_2)$

Meeting at Anywhere on Track

$\boxed{\text{Same dirn}}$ $\boxed{\text{oppo}}$

$t = \frac{L}{S_1 - S_2}$ $t = \frac{L}{S_1 + S_2}$

Circular Races :

Points to Remember :

1. When two runners have started at same time, from same point & they are running in same direction, then time required to meet for the first time at the starting point,

   LCM of (T_1, T_2)

2. When two runners have started at same time, from same point & they are running in opposite direction, then time required to meet for the first time at the starting point,

   LCM of (T1, T2)

Circular Races :

Points to Remember :

1. When two runners have started at same time, from same point & they are running in same direction, then during their first meeting, the faster runner always cover 1 round more than slower runner.

2. When two runners have started at same time, from same point & they are running in opposite direction, then during their first meeting, the faster & slower runner together cover 1 round.

3. When two runners have started at same time, from same point & they are running in same direction, then the time required for their first meeting,

$$t = \frac{L}{Sf - Ss}$$

4. When two runners have started at same time, from same point & they are running in opposite direction, then the time required for their first meeting,

$$t = \frac{L}{Sf + Ss}$$

QUESTION- 3

$K < A$ $S_K = 12$ km/h

#Q. Karan & Arjun started a walkathon around a circular track starting from the same point on the track in opposite direction. They met for the first after time 't'. Had they walked in the same direction with their speeds intact, they would have met after a time '7t'. It was also observed that Karan was slower than Arjun, and Karan's speed was measured as 12 kmph. Find the speed of Arjun (in m/s).

oppo. dirⁿ

$$\text{Time} = \frac{\text{dist}}{S_R}$$

$$t = \frac{L}{S_K + S_A}$$

$$t(S_K + S_A) = L$$

Same dirⁿ

$$\text{Time} = \frac{\text{dist}}{S_R}$$

$$7t = \frac{L}{S_A - S_K}$$

$$7t(S_A - S_K) = L$$

$$\cancel{t}(S_A + S_K) = 7\cancel{t}(S_A - S_K)$$

$$S_A + 12 = 7(S_A - 12)$$

$$S_A + 12 = 7S_A - 84$$

$$96 = 6S_A$$

$$\boxed{16 = S_A}$$

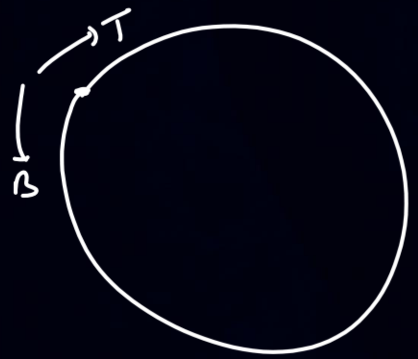
km/h

$$S_A = 16 \times \frac{5}{18} = \frac{80}{18} \text{ m/s} = 5.444 \text{ m/s}$$

QUESTION- 4

$L = 40m$

#Q. Tina and Bina jog in a circular track of length 40m. They started from the same starting point at the same time and met for the first time after 8 seconds, if Tina and Bina move in the opposite directions at speeds in the ratio of 3 : 2 respectively. Then find when they will meet for the 10th time had they moved in the same direction?



A 350

B 400

C 450

D 500

$\boxed{\text{Oppo. dir}}$

$t = 8$ sec

$t = \frac{L}{S_T + S_B}$

$8 = \frac{40}{3x+2x} = \frac{40}{5x}$

$40x = 40$

$\boxed{x = 1}$

$\boxed{\text{Same}}$

$t = \frac{L}{S_T - S_B}$

$= \frac{40}{3-2}$

$\boxed{t = 40 \text{ sec}}$

$\therefore$ time for 10th meeting

$= 10 \times 40$

$= 400$ sec

$\boxed{\begin{matrix} S_T : S_B \\ 3 : 2 \end{matrix}}$

$S_T = 3x$ m/s

$S_B = 2x$ m/s

$S_T = 3(1) = 3$ m/s

$S_B = 2(1) = 2$ m/s

QUESTION- 5

$\boxed{L = 3\text{ km}}$

#Q. Anil, Sunil, and Ravi run along a circular path of length 3 km, starting from the same point at the same time, and going in the clockwise direction. If they run at speeds of 15 km/hr, 10 km/hr and 8 km/hr, respectively, how much distance in km will Ravi have run when Anil and Sunil meet again for the first time at the starting point? [CAT 2020 : Slot 3]

(same)

$\frac{36}{60}$ h = 0.6 h

A) 4.6

B) 4.2

C) 4.8

D) 5.2

$S_A = 15$ km/h

$S_S = 10$ km/h

$S_R = 8$ km/h

$T_A \rightarrow \frac{L}{S_A} = \frac{3}{15} = 0.2 \xrightarrow{\times 60} \boxed{12\text{ min}}$

$T_S \rightarrow \frac{L}{S_S} = \frac{3}{10} = 0.3 \xrightarrow{\times 60} \boxed{18\text{ min}}$

A & S : meet for the 1st time at Starting Point after.

$t = \text{LCM}(T_A, T_S)$

$= \text{LCM}(0.3, 0.2)$

$\boxed{t = 0.6\text{ hr}}$

$\boxed{R}$

$d = S_R \times t$

$= 8 \times 0.6$

$= 4.8$ km

Averages

Simple Averages : $(\bar{x})$

$$\boxed{\bar{X} = \frac{\Sigma x}{N}}$$

$$\boxed{10, 15, 20, 25, 28}$$

$$\bar{X} = \frac{10+15+20+25+28}{5}$$

$$= \boxed{19.6}$$

$$\boxed{\bar{X} = \frac{\Sigma x}{N}}$$

$$\boxed{N = \frac{\Sigma x}{\bar{X}}}$$

$$\boxed{N \cdot \bar{X} = \Sigma x}$$

$N = 35$

$Avg = 40$

$$\Sigma x = N . \bar{X}$$

$$= 35 \times 40$$

$$= \underline{1400}$$

Combined Averages :

$N = 100$

$N_1 = 30$, $\bar{x}_1 = 100$

$N_2 = 25$, $\bar{x}_2 = 300$

$N_3 = 45$, $\bar{x}_3 = 200$

$$CA = \frac{N_1\bar{x}_1 + N_2\bar{x}_2 + N_3\bar{x}_3}{N_1 + N_2 + N_3}$$

$$= \frac{(30\times100)+(25\times300)+(45\times200)}{30+25+45}$$

$$= \frac{3000+7500+9000}{100} = \frac{195\cancel{00}}{1\cancel{00}}$$

$$= \boxed{195}$$

QUESTION- 6

#Q. The average of three integers is 13 . When a natural number n is included, the average of these four integers remains an odd integer. The minimum possible value of n is $\boxed{5}$ [CAT 2022 : Slot 1]

$$\frac{I + II + III}{3} = 13$$

$$\boxed{I + II + III = 39}$$

$$\frac{I + II + III + n}{4} \rightarrow \text{odd integer} \qquad n \rightarrow 1, 2, 3 \ldots$$

$$\frac{39 + n}{4} \rightarrow \boxed{\text{odd integer}}$$

$\boxed{n = 1}$: $\frac{40}{4} = 10$ (even)

$\boxed{n = 2}$: $\frac{41}{4} \neq$ integer

$\boxed{n = 3}$: $\frac{42}{4} \neq$ integer

$\boxed{n = 4}$: $\frac{43}{4} \neq$ integer

$\boxed{n = 5}$: $\frac{44}{4} = \boxed{11}$

(A) 3

(B) 4

(C) 5

(D) 1

QUESTION- 7

#Q. A family consists of two grandparents, two parents and three grandchildren. The average age of the grandparents is 67 years, that of the parents is 35 years and that of the grandchildren is 6 years. What is the average age of the family?

$31\left(\frac{5}{7}\right) = \frac{(31\times 7)+5}{7} = \frac{222}{7}$

CA = ?

GP: $N_1 = 2$, $\bar{X}_1 = 67$

P: $N_2 = 2$, $\bar{X}_2 = 35$

GC: $N_3 = 3$, $\bar{X}_3 = 6$

$CA = \frac{(2\times 67)+(2\times 35)+(3\times 6)}{2+2+3} = \frac{134+70+18}{7} = \boxed{\frac{222}{7}}$

(A) $28\frac{4}{7}$ years

(B) $31\frac{5}{7}$ years

(C) $32\frac{1}{7}$ years

(D) None of these

QUESTION- 8

#Q. If a certain amount of money is divided equally among n persons, each one receives Rs 352 . However, if two persons receive Rs 506 each and the remaining amount is divided equally among the other persons, each of them receive less than or equal to Rs 330 . Then, the maximum possible value of n is $\boxed{16}$

[CAT 2023 : Slot 2]

$\frac{\text{T. Amt}}{n} = 352$

$\boxed{\text{T. Amt} = 352n}$

I → 506

II → 506

1012

Remaining Amt = (352n − 1012)

Remaining People = (n − 2)

$\frac{\text{Remaining Amt}}{\text{Remaining People}} \leq 330$

$\frac{352n - 1012}{(n-2)} \leq 330$

$352n - 1012 \leq 330n - 660$

$22n \leq 1012 - 660$

$22n \leq 352$

$n \leq \frac{352}{22}$

$n \leq 16$

n → $\boxed{16}$, 15, 14 - - - .

QUESTION- 9

let, Avg. of A, B, C $\rightarrow$ y kg

#Q. There are three persons A, B and C in a room. If a person D joins the room, the average weight of the persons in the room reduces by x kg. Instead of D, if person E joins the room, the average weight of the persons in the room increases by 2x kg. If the weight of E is 12 kg more than that of D, then the value of x is

[CAT 2023 : Slot 3]

$$\frac{A+B+C}{3} = y$$

$$\boxed{A+B+C = 3y}$$

$$\frac{A+B+C+D}{4} = (y-x)$$

$$3y + D = 4y - 4x$$

$$\boxed{D = y - 4x}$$

$$\frac{A+B+C+E}{4} = (y+2x)$$

$$3y + E = 4y + 8x$$

$$\boxed{E = y + 8x}$$

$$E = D + 12$$

$$\cancel{y} + 8x = \cancel{y} - 4x + 12$$

$$12x = 12$$

$$\boxed{x = 1}$$

(A) 1.5

(B) 2

(C) 0.5

(D) 1 ✓

QUESTION- 10

let I(a) → 1

#Q. There are four numbers such that average of first two numbers is 1 more than the first number, average of first three numbers is 2 more than average of first two numbers, and average of first four numbers is 3 more than average of first three numbers. Then, the difference between the largest and the smallest numbers, is

[CAT 2024 : Slot 1]

$\boxed{15}$

$\frac{a+b}{2} = 4 \Rightarrow b = 5$

(a)	(b)	(c)	(d)
I	II	III	IV
1	3	8	16

diff = 15

$\frac{a+b}{2} = 1+1$

$\boxed{a+b=4}$

$1+b=4$

$\boxed{b=3}$

$\frac{a+b+c}{3} = 2+2$

$4+c=12$

$\boxed{c=8}$

$\frac{a+b+c+d}{4} = 4+3$

$1+3+8+d=28$

$\boxed{d=16}$

QUESTION- 11

Simplest Avg $\Rightarrow$ Arithmetic Mean

#Q. The average of three distinct real numbers is 28. If the smallest number is increased by 7 and the largest number is reduced by 10, the order of the numbers remains unchanged, and the new arithmetic mean becomes 2 more than the middle number, while the difference between the largest and the smallest numbers becomes 64. Then, the largest number in the original set of three numbers is

[CAT 2024 : Slot 1]

I $\rightarrow$ a (Smallest)
II $\rightarrow$ b (Middle)
III $\rightarrow$ c (Largest)

$\therefore c = 81 + a$
$= 81 + (-11)$
$= 70$

$\frac{a+b+c}{3} = 28$

$\boxed{a+b+c = 84}$

$a + 25 + 81 + a = 84$

$2a = -22$

$\boxed{a = -11}$

New

(s) I $\rightarrow$ $(a+7)$
(m) II $\rightarrow$ b
(l) III $\rightarrow$ $(c-10)$

III $-$ I $= 64$

$c - 10 - (a+7) = 64$

$c - 10 - a - 7 = 64$

$c - a = 81$

$\boxed{c = 81 + a}$

$\frac{a+7+b+c-10}{3} = (b+2)$

$a+b+c-3 = 3b+6$

$84 - 3 = 3b + 6$

$75 = 3b$

$\boxed{25 = b}$

2 Mins Summary

1) Concepts & Numerical based on ‘Linear Races’

2) Concepts & Numerical based on ‘Circular Races’

3) Concept & Numerical based on ‘Averages’

4) Problems for Practice & CAT PYQs

Thank You